F79SP Session 11: Recurrence and Transience

We are interested in the property of whether a state will always be revisited, and will then relate this property to the notion of open and closed classes that we studied before. Proofs will be covered in a subsequent session.

1 Recurrent and transient states

Define $F_{ii} = P(\text{MC } \{X_n\} \text{ ever returns to state } i \mid X_0 = i)$.

Definition 1.1. *A state $i \in S$ is called:*

- *a **recurrent state** if $F_{ii} = 1$,*
- *otherwise, a **transient state** (if $F_{ii} < 1$).*

In words, recurrent state can be revisited with probability 1, whereas there is a non-zero probability that a transient state cannot be revisited.

Example 1.1. *Consider a THMC with finite state space $S = \{0, 1, 2, 3\}$ and transition matrix:*

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.7 & 0 & 0.3 & 0 \\ 0 & 0.7 & 0 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Observe that: We can further see that:

- *States in the same class has the same transience and recurrence property.*
- *Given starting at State 0, the expected time to first return to itself is finite (exactly one time-step).*

The following result provides some further interpretations to the notion of recurrence and transience.

Theorem 1.1. *For any state $i \in \mathcal{S}$, we have*

$$\begin{aligned} i \text{ recurrent} &\Leftrightarrow F_{ii} = 1 \\ &\Leftrightarrow \mathbf{P}(\text{process } \{X_n\} \text{ ever returns to state } i \,|\, X_0 = i) = 1 \\ &\Leftrightarrow \mathbf{P}(\text{process } \{X_n\} \text{ returns to state } i \text{ infinitely often} \,|\, X_0 = i) = 1 \\ &\Leftrightarrow \mathbf{E}\,(\text{number of returns by process } \{X_n\} \text{ to state } i \,|\, X_0 = i) = \infty \\ &\Leftrightarrow \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

and also

$$\begin{aligned} i \text{ transient} &\Leftrightarrow F_{ii} < 1 \\ &\Leftrightarrow \mathbf{P}(\text{process } \{X_n\} \text{ ever returns to state } i \,|\, X_0 = i) < 1 \\ &\Leftrightarrow \mathbf{P}(\text{process } \{X_n\} \text{ returns to state } i \text{ infinitely often} \,|\, X_0 = i) = 0 \\ &\Leftrightarrow \mathbf{E}\,(\text{number of returns by process } \{X_n\} \text{ to state } i \,|\, X_0 = i) < \infty \\ &\Leftrightarrow \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty. \end{aligned}$$

Some remarks are given as follows:

1. Due to Markov property, returning once with probability 1 implies that returning infinitely often with probability 1.
2. It can be shown that $\mathbf{E}\,(\text{number of returns by process } \{X_n\} \text{ to state } i \,|\, X_0 = i) = \sum_{n=1}^{\infty} p_{ii}^{(n)}$.

We thus have another interpretation of transience and recurrence as follows:

- Starting from a **transient** state, the MC $\{X_n\}$ eventually leaves it forever.
- Starting from a **recurrent** state, the MC $\{X_n\}$ will revisit it infinitely often.

The following corollary is useful for justifying whether a state is recurrent (using its contrapositive).

Corollary 1.2. *If $i \in \mathcal{S}$ is **transient**, then $\lim_{n\to\infty} p_{ii}^{(n)} = 0$.*

Proof. If $i \in \mathcal{S}$ is transient, then $\sum_{n=1}^{\infty} p_{ii}^{(n)}$ converges, then $\lim_{n\to\infty} p_{ii}^{(n)} = 0$ by the Divergence Test. □

Example 1.2. *From Example 1.1, note that $p_{00} = 1$ and thus $p_{00}^{(n)} = 1$ for any n.*

Recurrent states can be further classified based on its first returning time:

$$\mu_i = \mathbf{E}\,(\text{time to first return to state } i \,|\, X_0 = i).$$

Definition 1.2. *A recurrent state $i \in \mathcal{S}$ is **positive recurrent** if $\mu_i < \infty$ and **null recurrent** if $\mu_i = \infty$. Here, $1/\mu_i$ is the intensity to return to i.*

2 Recurrent and transient classes

The idea of recurrence and transience can be extended from states to classes.

Theorem 2.1. *In any class (of intercommunicating states), either all states are recurrent, or all states are transient. Hence, we can call any class as either a **recurrent class** or a **transient class**.*

Note: Given a class. If we know one of its states is transient (recurrent), then every state in the class must also be transient (recurrent).

We illustrate the connections between recurrent/transient class and closed/open classes.

Result 2.1. *For any class of intercommunicating states:*

- *recurrent class* $\Longrightarrow$

- *open class* $\Longrightarrow$

- *finite closed class* $\Longrightarrow$

- *For finite state space* S*,*

Example 2.1. *Reconsider the THMC in Example 1.1. Alternatively, since* S *is finite, we have both closed classes* $\{0\}, \{3\}$ *are recurrent, and the open class* $\{1, 2\}$ *is transient.*

Bonus-malus system. Recall that, provided $0 < p < 1$, the state space $\mathcal{S}$ is here a single, necessarily closed, class. Hence it is recurrent.

Health insurance system. Recall that here we obtained the earlier classification:

Class	Type
$\{H, S\}$	nonclosed
$\{D\}$	closed (absorbing)

It follows that the states H and S (which form a nonclosed class) are transient, while the state D (which forms a finite closed class) is recurrent.

Simple random walk. For the simple random walk with $0 < p < 1$ (and $q = 1 - p$), we have already observed that the state space is a single, necessarily closed, class. To check recurrence/transience, it is sufficient to compute F_{00}. Define $F_{ij} = P(\text{MC } \{X_n\} \text{ hits state } j \text{ at some time } n > 0 \mid X_0 = i)$.

Solution 1. By conditioning as usual on the first step of the random walk, we have

$$F_{00} = pF_{10} + qF_{-10}$$

- For $p > 1/2$, we can use the result from Ruin problem to conclude that $F_{10} = q/p$ and $F_{-10} = 1$. Hence $F_{00} = 2q < 1$ and so state 0 (and hence all states) are transient.
- For $p = 1/2$, we have $F_{10} = 1$ and $F_{-10} = 1$. Hence $F_{00} = 1$ and so state 0 (and hence all states) are recurrent.
- For $p < 1/2$, it follows by the result from Ruin problem and by symmetry (interchanging p and q) that $F_{00} = 2p < 1$ and so state 0 (and hence all states) are again transient.

Solution 2.

- By Theorem 1.1, state 0 is recurrent if and only if $\sum_{n=1}^{\infty} p_{00}^{(n)} = \infty$.
- Earlier, using Stirling's approximation, we obtained that $p_{00}^{(2n)} \sim \frac{1}{\sqrt{2\pi n}} \cdot (4pq)^n$ and that $p_{00}^{(2n+1)} = 0$.
- Then, by basic results from calculus, $\sum p_{00}^{(n)} = \infty$ if and only if $4pq = 1$, and if and only if $p = q = 1/2$.

Exercise. Show that in the simple random walk on the nonnegative integers, modified so that state 0 is an absorbing state, this latter state is recurrent while all other states are transient.